

Concept 2 | Triangles and Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Line

$$y - y_1 = m(x - x_1)$$

Perpendicular

$$m_1 m_2 = -1$$

Area

$$A = \frac{1}{2}bh$$

Classified Practice

Triangles and Lines

Q001 Written

Determine the shape of triangle ABC with A (3, 2), B (−1, 0), C $(1 + \sqrt{3}, 1 - 2\sqrt{3})$.

Classified Practice

Triangles and Lines

Q002 **Written**

Determine the shape of triangle ABC with A (2, 2), B (2, 0), C $(2 + \sqrt{3}, 1)$.

Classified Practice

Triangles and Lines

Q003 MCQ

Determine the shape of triangle ABC with A (3, 2), B (−1, 0), C $(1 + \sqrt{3}, 1 - 2\sqrt{3})$:

Choose the correct option.

A *isosceles (not right – angled)*

B *scalene (not right – angled)*

C *scalene right – angled*

D *equilateral*

Classified Practice

Triangles and Lines

Q004 MCQ

Determine the shape of triangle ABC with A (2, 2), B (2, 0), C $(2 + \sqrt{3}, 1)$:

Choose the correct option.

- A *scalene (not right – angled)*
- B *scalene right – angled*
- C *equilateral*
- D *isosceles (not right – angled)*

Q005 Written**Determine the shape of triangle ABC with A $(-1, 2)$, B $(1, 1)$, C $(0, 4)$.**

Classified Practice

Triangles and Lines

Q006 Written

Determine the shape of triangle ABC with A (0, 2), B (−1, −2), C (4, 1).

Q007 MCQ**Determine the shape of triangle ABC with A $(-1, 2)$, B $(1, 1)$, C $(0, 4)$:****Choose the correct option.****A** *scalene right – angled***B** *isosceles right – angled; right angle at A***C** *equilateral***D** *isosceles (not right – angled)*

Classified Practice

Triangles and Lines

Q008 MCQ

Determine the shape of triangle ABC with A (0, 2), B (−1, −2), C (4, 1):

Choose the correct option.

A *equilateral*

B *isosceles (not right – angled)*

C *scalene right – angled*

D *isosceles right – angled; right angle at A*

Q009 Written**Determine the shape of triangle ABC with A (−3, 4), B (1, 7), C (3, −4).**

Classified Practice

Triangles and Lines

Q010 MCQ

Determine the shape of triangle ABC with A (−3, 4), B (1, 7), C (3, −4):

Choose the correct option.

A *equilateral*

B *isosceles (not right – angled)*

C *scalene (not right – angled)*

D *scalene right – angled; right angle at A*

Q011 Written**Determine the shape of triangle ABC with A (4, 9), B (0, 6), C (3, 2).**

Classified Practice

Triangles and Lines

Q012 **Written****Determine the shape of triangle ABC with A (1, 2), B (5, 8), C (11, 4).**

Q013 Written

Determine the shape of triangle ABC with A (−1, 1), B (4, 2), C (1, 4).

Classified Practice

Triangles and Lines

Q014 MCQ

Determine the shape of triangle ABC with A (4, 9), B (0, 6), C (3, 2):**Choose the correct option.**A *scalene right – angled*B *isosceles right – angled; right angle at B*C *equilateral*D *isosceles (not right – angled)*

Q015 MCQ**Determine the shape of triangle ABC with A (1, 2), B (5, 8), C (11, 4):****Choose the correct option.****A** *isosceles (not right – angled)***B** *scalene right – angled***C** *isosceles right – angled; right angle at B***D** *equilateral*

Classified Practice

Triangles and Lines

Q016 MCQ

Determine the shape of triangle ABC with A $(-1, 1)$, B $(4, 2)$, C $(1, 4)$:

Choose the correct option.

A *scalene right – angled*

B *isosceles right – angled; right angle at C*

C *equilateral*

D *isosceles (not right – angled)*

Classified Practice

Triangles and Lines

Q017 Challenge

If A (1, 2), B (2k, k), C (4, 5) and ABC is isosceles and right-angled at B, find k.

Classified Practice

Triangles and Lines

Q018 Challenge

If A (1, 2), B (2k, k), C (4, 5) and ABC is isosceles and right-angled at B, find k:

Choose the correct option.

A $k = 1$

B $k = -2$

C $k = 2$

D $k = 3$

Classified Practice

Triangles and Lines

Q019 Written

Find the equation of the line through (3,-4) parallel to the x-axis.

Q020 Written**Find the equation of the line through $(-3,8)$ parallel to the y-axis.**

Classified Practice

Triangles and Lines

Q021 **Written****Find the equation of the line through (2,6) parallel to the x-axis.**

Q022 Written**Find the equation of the line through (3,5) perpendicular to the x-axis.**

Classified Practice

Triangles and Lines

Q023 **Written****Find the equation of the line through $(-4,5)$ parallel to x-axis.**

Q024 Written**Find the equation of the line through (2,1) parallel to x-axis.**

Classified Practice

Triangles and Lines

Q025 Written

Find the equation of the line through $(-4, -7)$ parallel to x-axis.

Q026 Written**Find the equation of the line through (1,-2) perpendicular to x-axis.**

Classified Practice

Triangles and Lines

Q027 Written

Find the equation of the line through $(-3, -4)$ perpendicular to x-axis.

Q028 Written**Find the equation of the line through (3,-5) perpendicular to x-axis.**

Classified Practice

Triangles and Lines

Q029 Written

Find the equation of the line through (2,-7) parallel to y-axis.

Q030 Written**Find the equation of the line through (4,3) parallel to y-axis.**

Classified Practice

Triangles and Lines

Q031 **Written****Find the equation of the line through $(-6, -2)$ parallel to y-axis.**

Q032 MCQ**Find the equation of the line through (3,-4) parallel to the x-axis:****Choose the correct option.**

A $y = 4$

B $y = -4$

C $y = -3$

D $y = -5$

Classified Practice

Triangles and Lines

Q033 MCQ

Find the equation of the line through $(-3,8)$ parallel to the y-axis:**Choose the correct option.**

A $x = -4$

B $x = 3$

C $x = -3$

D $x = -2$

Q034 MCQ**Find the equation of the line through (2,6) parallel to the x-axis:****Choose the correct option.**

A $y = 7$

B $y = 5$

C $y = -6$

D $y = 6$

Classified Practice

Triangles and Lines

Q035 MCQ

Find the equation of the line through (3,5) perpendicular to the x-axis:**Choose the correct option.**

A $x = 4$

B $x = 2$

C $x = -3$

D $x = 3$

Q036 MCQ**Find the equation of the line through $(-4,5)$ parallel to x-axis:****Choose the correct option.**

A $y = -5$

B $y = 5$

C $y = 6$

D $y = 4$

Classified Practice

Triangles and Lines

Q037 MCQ

Find the equation of the line through (2,1) parallel to x-axis:**Choose the correct option.**

A $y = 2$

B $y = 0$

C $y = -1$

D $y = 1$

Q038 MCQ**Find the equation of the line through $(-4, -7)$ parallel to x-axis:****Choose the correct option.**

A $y = 7$

B $y = -7$

C $y = -6$

D $y = -8$

Classified Practice

Triangles and Lines

Q039 MCQ

Find the equation of the line through (1,-2) perpendicular to x-axis:**Choose the correct option.**

A $x = 0$

B $x = -1$

C $x = 1$

D $x = 2$

Q040 MCQ**Find the equation of the line through (-3,-4) perpendicular to x-axis:****Choose the correct option.**

A $x = -2$

B $x = -4$

C $x = 3$

D $x = -3$

Classified Practice

Triangles and Lines

Q041 MCQ

Find the equation of the line through (3,-5) perpendicular to x-axis:**Choose the correct option.**

A $x = 4$

B $x = 2$

C $x = -3$

D $x = 3$

Q042 MCQ**Find the equation of the line through (2,-7) parallel to y-axis:****Choose the correct option.**

A $x = -2$

B $x = 2$

C $x = 3$

D $x = 1$

Classified Practice

Triangles and Lines

Q043 MCQ

Find the equation of the line through (4,3) parallel to y-axis:**Choose the correct option.**

A $x = 5$

B $x = 3$

C $x = -4$

D $x = 4$

Q044 MCQ**Find the equation of the line through $(-6, -2)$ parallel to y -axis:****Choose the correct option.**

A $x = -6$

B $x = -5$

C $x = -7$

D $x = 6$

Classified Practice

Triangles and Lines

Q045 **Written****Find the equation of the line through $(-3,1)$ and $(-3,2)$.**

Q046 Written**Find the equation of the line through $(-2,3)$ and $(-5,3)$.**

Classified Practice

Triangles and Lines

Q047 **Written****Find the equation of the line through (1,-1) and (4,0).**

Q048 Written**Find the equation of the line through $(-2,-2)$ and $(2,-2)$.**

Classified Practice

Triangles and Lines

Q049 **Written****Find the equation of the line through $(-4,2)$ and $(-4,8)$.**

Q050 Written**Find the equation of the line through $(-2,-6)$ and $(-4,1)$.**

Classified Practice

Triangles and Lines

Q051 **Written****Find the equation of the line through (4,0) and (4,3).**

Q052 Written**Find the equation of the line through $(-1,2)$ and $(-1,6)$.**

Classified Practice

Triangles and Lines

Q053 **Written****Find the equation of the line through $(-5,-3)$ and $(-5,2)$.**

Q054 Written**Find the equation of the line through $(-2,3)$ and $(-3,3)$.**

Classified Practice

Triangles and Lines

Q055 **Written****Find the equation of the line through (4,-2) and (1,-2).**

Q056 Written**Find the equation of the line through $(-3,6)$ and $(2,6)$.**

Classified Practice

Triangles and Lines

Q057 **Written****Find the equation of the line through (1,2) and (3,10).**

Q058 Written**Find the equation of the line through $(-1,-8)$ and $(4,2)$.**

Classified Practice

Triangles and Lines

Q059 MCQ

Find the equation of the line through (-3,1) and (-3,2):**Choose the correct option.**

A $x = 3$

B $x = -3$

C $x = -2$

D $x = -4$

Q060 MCQ**Find the equation of the line through $(-2,3)$ and $(-5,3)$:****Choose the correct option.**

A $y = 2$

B $y = -3$

C $y = 3$

D $y = 4$

Classified Practice

Triangles and Lines

Q061 MCQ

Find the equation of the line through (1,-1) and (4,0):**Choose the correct option.**

A $y = -x/3 - 4/3$

B $y = 4/3 - x/3$

C $y = x/3 - 4/3 + 1$

D $y = x/3 - 4/3$ (equivalently $x - 3y - 4 = 0$)

Q062 MCQ**Find the equation of the line through (-2,-2) and (2,-2):****Choose the correct option.**

A $y = -2$

B $y = -1$

C $y = -3$

D $y = 2$

Classified Practice

Triangles and Lines

Q063 MCQ

Find the equation of the line through $(-4,2)$ and $(-4,8)$:**Choose the correct option.**

A $x = 4$

B $x = -4$

C $x = -3$

D $x = -5$

Q064 MCQ**Find the equation of the line through (-2,-6) and (-4,1):****Choose the correct option.**

A $7x + 2y + 26 + 1 = 0$

B $7x - 2y + 26 = 0$

C $2(7x + 2y + 26) + 1 = 0$

D $7x + 2y + 26 = 0$

Classified Practice

Triangles and Lines

Q065 MCQ

Find the equation of the line through (4,0) and (4,3):**Choose the correct option.**

A $x = 3$

B $x = -4$

C $x = 4$

D $x = 5$

Q066 MCQ**Find the equation of the line through $(-1,2)$ and $(-1,6)$:****Choose the correct option.**

A $x = -1$

B $x = 0$

C $x = -2$

D $x = 1$

Classified Practice

Triangles and Lines

Q067 MCQ

Find the equation of the line through $(-5,-3)$ and $(-5,2)$:**Choose the correct option.**

A $x = -6$

B $x = 5$

C $x = -5$

D $x = -4$

Q068 MCQ**Find the equation of the line through $(-2,3)$ and $(-3,3)$:****Choose the correct option.**

A $y = 4$

B $y = 2$

C $y = -3$

D $y = 3$

Classified Practice

Triangles and Lines

Q069 MCQ

Find the equation of the line through (4,-2) and (1,-2):**Choose the correct option.**

A $y = -1$

B $y = -3$

C $y = 2$

D $y = -2$

Q070 MCQ**Find the equation of the line through (-3,6) and (2,6):****Choose the correct option.**

A $y = 5$

B $y = -6$

C $y = 6$

D $y = 7$

Classified Practice

Triangles and Lines

Q071 MCQ

Find the equation of the line through (1,2) and (3,10):**Choose the correct option.**

A $y = -4x - 2$

B $y = 4x - 2 + 1$

C 0

D $y = 4x - 2$

Q072 MCQ**Find the equation of the line through (-1,-8) and (4,2):****Choose the correct option.**

A $y = -2x - 6$

B $y = 2x - 6 + 1$

C 0

D $y = 2x - 6$

Classified Practice

Triangles and Lines

Q073 **Written****Find the equation of the line through (2,-1) with gradient 3.**

Q074 Written**Find the equation of the line through $(-3,4)$ with gradient 2.**

Classified Practice

Triangles and Lines

Q075 **Written****Find the equation of the line through $(-1, 5)$ with gradient -1 .**

Q076 Written**Find the equation of the line through $(-3, -8)$ with gradient 1.**

Classified Practice

Triangles and Lines

Q077 Written

Find the equation of the line through (2,-1) with gradient -2.

Q078 MCQ**Find the equation of the line through (2,-1) with gradient 3:****Choose the correct option.**

A $y = -3x - 7$

B $y = 3x - 7 + 1$

C 0

D $y = 3x - 7$

Classified Practice

Triangles and Lines

Q079 MCQ

Find the equation of the line through (-3,4) with gradient 2:**Choose the correct option.**

A $y = 2x + 10 + 1$

B 0

C $y = 2x + 10$

D $y = -2x + 10$

Q080 MCQ**Find the equation of the line through (-1,5) with gradient -1:****Choose the correct option.**

A $y = x + 4$

B $y = x - 4$

C $y = -x + 4 + 1$

D $y = -x + 4$

Classified Practice

Triangles and Lines

Q081 MCQ

Find the equation of the line through (-3,-8) with gradient 1:**Choose the correct option.**

A $y = x - 5 + 1$

B $y = x - 5$

C $y = -x - 5$

D $y = 5 - x$

Q082 MCQ**Find the equation of the line through (2,-1) with gradient -2:****Choose the correct option.**

A $y = 2x + 3$

B $y = -2x + 3 + 1$

C 0

D $y = -2x + 3$

Classified Practice

Triangles and Lines

Q083 **Challenge**

Sensors A $(-4, 2)$ and B $(9, 2)$ have the same height. Find the straight-line boundary through A and B.

Classified Practice

Triangles and Lines

Q084 Challenge

Sensors A $(-4, 2)$ and B $(9, 2)$ have the same height. Find the straight-line boundary through A and B:

Choose the correct option.

A $y = 3$

B $y = 1$

C $y = -2$

D $y = 2$

Classified Practice

Triangles and Lines

Q085 Written

A $(2, 3)$, **B** $(4, -1)$, and **C** $(a - 1, 2 - a)$ are collinear. Find **a**.

Q086 Written

A (2, 5), B (4, 9), and C (-1, a) are collinear. Find a.

Classified Practice

Triangles and Lines

Q087 Written

A (1, -1), **B** (-2, -3), and **C** (4, a) are collinear. Find a .

Q088 Written

A (1, 2), **B** (3, 1), and **C** (a , -1) are collinear. Find a .

Classified Practice

Triangles and Lines

Q089 Written

A $(-3, -9)$, **B** $(5, 7)$, and **C** $(a + 3, 3a - 2)$ are collinear. Find **a**.

Q090 MCQ**A** $(2, 3)$, **B** $(4, -1)$, and **C** $(a - 1, 2 - a)$ are collinear. Find a :**Choose the correct option.**

A $a = 8$

B $a = 6$

C $a = -7$

D $a = 7$

Classified Practice

Triangles and Lines

Q091 MCQ

A (2, 5), **B** (4, 9), and **C** (−1, a) are collinear. Find a :

Choose the correct option.

A $a = 0$

B $a = -2$

C $a = 1$

D $a = -1$

Q092 MCQ**A (1, -1), B (-2, -3), and C (4, a) are collinear. Find a:****Choose the correct option.****A $a = 0$** **B $a = -1$** **C $a = 1$** **D $a = 2$**

Classified Practice

Triangles and Lines

Q093 MCQ

A (1, 2), **B** (3, 1), and **C** (a , -1) are collinear. Find a :

Choose the correct option.

A $a = 8$

B $a = 6$

C $a = -7$

D $a = 7$

Q094 MCQ

A $(-3, -9)$, **B** $(5, 7)$, and **C** $(a + 3, 3a - 2)$ are collinear. Find a :

Choose the correct option.

A $a = 6$

B $a = 4$

C $a = -5$

D $a = 5$

Classified Practice

Triangles and Lines

Q095 Written

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through $(-1, 3)$.

Q096 Written**Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through (5,6).**

Classified Practice

Triangles and Lines

Q097 Written

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through $(-1, 2)$.

Q098 Written

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through $(-3, -1)$.

Classified Practice

Triangles and Lines

Q099 MCQ

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through $(-1,3)$:

Choose the correct option.

A $2(x + y - 2) + 1 = 0$

B $x + y - 2 = 0$

C $x + y - 2 + 1 = 0$

D $x - y - 2 = 0$

Q100 MCQ

Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through (5,6):

Choose the correct option.

A $x + 3y - 23 = 0$

B $x + 3y - 23 + 1 = 0$

C $x - 3y - 23 = 0$

D $2(x + 3y - 23) + 1 = 0$

Classified Practice

Triangles and Lines

Q101 MCQ

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through $(-1, 2)$:

Choose the correct option.

A $2(x + y - 1) + 1 = 0$

B $x + y - 1 = 0$

C $x + y - 1 + 1 = 0$

D $x - y - 1 = 0$

Q102 MCQ

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through $(-3, -1)$:

Choose the correct option.

A $2x - 5y + 1 + 1 = 0$

B $2x - 5y - 1 = 0$

C $2(2x - 5y + 1) + 1 = 0$

D $2x - 5y + 1 = 0$

Classified Practice

Triangles and Lines

Q103 Challenge

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory.

Classified Practice

Triangles and Lines

Q104 Challenge

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory:

Choose the correct option.

A $3x - 13y = 0$

B $2(3x + 13y) + 1 = 0$

C $3x + 13y = 0$

D $3x + 13y + 1 = 0$

Classified Practice

Triangles and Lines

Q105 Written

Find the gradient of line $y = 2x + 3$.

Q106 Written**Find the gradient of line $y = -3x + 2$.**

Classified Practice

Triangles and Lines

Q107 **Written****Find the gradient of line $x - 3y + 1 = 0$.**

Q108 Written**Find the gradient of line $2x - y - 1 = 0$.**

Classified Practice

Triangles and Lines

Q109 Written

Find the gradient of line $x + 3y + 3 = 0$.

Q110 Written**Find the gradient of line $y = 3x + 1$.**

Classified Practice

Triangles and Lines

Q111 Written

Find the gradient of line $y = 4x - 1$.

Q112 Written**Find the gradient of line $y = -4x - 7$.**

Classified Practice

Triangles and Lines

Q113 Written

Find the gradient of line $6x - 2y = 5$.

Q114 Written**Find the gradient of line $8x + 2y - 3 = 0$.**

Classified Practice

Triangles and Lines

Q115 **Written****Find the gradient of line $x + 4y - 4 = 0$.**

Q116 Written**Find the gradient of line $y = 3x - 1$.**

Classified Practice

Triangles and Lines

Q117 **Written****Find the gradient of line $y = -3x + 4$.**

Q118 Written**Find the gradient of line $y = x - 3$.**

Classified Practice

Triangles and Lines

Q119 Written

Find the gradient of line $6x + 2y + 5 = 0$.

Q120 Written**Find the gradient of line $2x - 2y - 1 = 0$.**

Classified Practice

Triangles and Lines

Q121 **Written****Find the gradient of line $x + 3y - 2 = 0$.**

Q122 Written**Find the gradient of line $y = 2x - 1$.**

Classified Practice

Triangles and Lines

Q123 Written

Find the gradient of line $y = -x + 4$.

Q124 Written**Find the gradient of line $y = 3x/2 + 6$.**

Classified Practice

Triangles and Lines

Q125 Written

Find the gradient of line $x + y + 3 = 0$.

Q126 Written**Find the gradient of line $2x + 3y + 8 = 0$.**

Classified Practice

Triangles and Lines

Q127 **Written****Find the gradient of line $3x + 6y - 1 = 0$.**

Q128 MCQ

Find the gradient of line $y = 2x + 3$:**Choose the correct option.**

A $m = 3$

B $m = 1$

C $m = -2$

D $m = 2$

Classified Practice

Triangles and Lines

Q129 MCQ

Find the gradient of line $y = -3x + 2$:**Choose the correct option.**A $m = -4$ B $m = 3$ C $m = -3$ D $m = -2$

Q130 MCQ**Find the gradient of line $x - 3y + 1 = 0$:****Choose the correct option.**

A $m = 1/3$

B $m = 4/3$

C $m = -2/3$

D $m = -1/3$

Classified Practice

Triangles and Lines

Q131 MCQ

Find the gradient of line $2x - y - 1 = 0$:**Choose the correct option.**A $m = -2$ B $m = 2$ C $m = 3$ D $m = 1$

Q132 MCQ**Find the gradient of line $x + 3y + 3 = 0$:****Choose the correct option.**

A $m = -1/3$

B $m = 2/3$

C $m = -4/3$

D $m = 1/3$

Classified Practice

Triangles and Lines

Q133 MCQ

Find the gradient of line $y = 3x + 1$:**Choose the correct option.**

A $m = 3$

B $m = 4$

C $m = 2$

D $m = -3$

Q134 MCQ**Find the gradient of line $y = 4x - 1$:****Choose the correct option.**

A $m = -4$

B $m = 4$

C $m = 5$

D $m = 3$

Classified Practice

Triangles and Lines

Q135 MCQ

Find the gradient of line $y = -4x - 7$:**Choose the correct option.**

A $m = -3$

B $m = -5$

C $m = 4$

D $m = -4$

Q136 MCQ**Find the gradient of line $6x - 2y = 5$:****Choose the correct option.**

A $m = 3$

B $m = 4$

C $m = 2$

D $m = -3$

Classified Practice

Triangles and Lines

Q137 MCQ

Find the gradient of line $8x + 2y - 3 = 0$:**Choose the correct option.**

A $m = -5$

B $m = 4$

C $m = -4$

D $m = -3$

Q138 MCQ**Find the gradient of line $x + 4y - 4 = 0$:****Choose the correct option.**

A $m = -5/4$

B $m = 1/4$

C $m = -1/4$

D $m = 3/4$

Classified Practice

Triangles and Lines

Q139 MCQ

Find the gradient of line $y = 3x - 1$:**Choose the correct option.**

A $m = 4$

B $m = 2$

C $m = -3$

D $m = 3$

Q140 MCQ**Find the gradient of line $y = -3x + 4$:****Choose the correct option.**

A $m = -3$

B $m = -2$

C $m = -4$

D $m = 3$

Classified Practice

Triangles and Lines

Q141 MCQ

Find the gradient of line $y = x - 3$:**Choose the correct option.**

A $m = 2$

B $m = 0$

C $m = -1$

D $m = 1$

Q142 MCQ**Find the gradient of line $6x + 2y + 5 = 0$:****Choose the correct option.**

A $m = -4$

B $m = 3$

C $m = -3$

D $m = -2$

Classified Practice

Triangles and Lines

Q143 MCQ

Find the gradient of line $2x - 2y - 1 = 0$:**Choose the correct option.**

A $m = 0$

B $m = -1$

C $m = 1$

D $m = 2$

Q144 MCQ**Find the gradient of line $x + 3y - 2 = 0$:****Choose the correct option.**

A $m = -1/3$

B $m = 2/3$

C $m = -4/3$

D $m = 1/3$

Classified Practice

Triangles and Lines

Q145 MCQ

Find the gradient of line $y = 2x - 1$:**Choose the correct option.**

A $m = 3$

B $m = 1$

C $m = -2$

D $m = 2$

Q146 MCQ**Find the gradient of line $y = -x + 4$:****Choose the correct option.**

A $m = -1$

B $m = 0$

C $m = -2$

D $m = 1$

Classified Practice

Triangles and Lines

Q147 MCQ

Find the gradient of line $y = 3x/2 + 6$:**Choose the correct option.**

A $m = -3/2$

B $m = 3/2$

C $m = 5/2$

D $m = 1/2$

Q148 MCQ**Find the gradient of line $x + y + 3 = 0$:****Choose the correct option.**

- A $m = 0$
- B $m = -2$
- C $m = 1$
- D $m = -1$

Classified Practice

Triangles and Lines

Q149 MCQ

Find the gradient of line $2x + 3y + 8 = 0$:**Choose the correct option.**

A $m = 2/3$

B $m = -2/3$

C $m = 1/3$

D $m = -5/3$

Q150 MCQ**Find the gradient of line $3x + 6y - 1 = 0$:****Choose the correct option.****A** 0**B** $m = -1/2$ **C** $m = 1/2$ **D** $m = -3/2$

Classified Practice

Triangles and Lines

Q151 **Written****Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$.**

Q152 Written

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$.

Classified Practice

Triangles and Lines

Q153 **Written****Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$.**

Q154 Written

Find the equation of the line through $(-1,2)$ parallel to $2x + 3y = 0$.

Classified Practice

Triangles and Lines

Q155 **Written**

Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$.

Q156 MCQ**Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$:****Choose the correct option.**

- A $3x + 2y - 14 = 0$
- B $2(3x - 2y - 14) + 1 = 0$
- C $3x - 2y - 14 = 0$
- D $3x - 2y - 14 + 1 = 0$

Classified Practice

Triangles and Lines

Q157 MCQ

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$:**Choose the correct option.**

- A $2x + y + 1 + 1 = 0$
B $2x - y + 1 = 0$
C $2(2x + y + 1) + 1 = 0$
D $2x + y + 1 = 0$
-
-
-

Q158 MCQ**Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$:****Choose the correct option.**

A $2x + y - 4 = 0$

B $2(2x - y - 4) + 1 = 0$

C $2x - y - 4 = 0$

D $2x - y - 4 + 1 = 0$

Classified Practice

Triangles and Lines

Q159 MCQ

Find the equation of the line through (-1,2) parallel to $2x + 3y = 0$:**Choose the correct option.**

A $2x + 3y - 4 + 1 = 0$

B $2x - 3y - 4 = 0$

C $2(2x + 3y - 4) + 1 = 0$

D $2x + 3y - 4 = 0$

Q160 MCQ

Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$:

Choose the correct option.

A $3x + 2y - 7 + 1 = 0$

B $3x - 2y - 7 = 0$

C $2(3x + 2y - 7) + 1 = 0$

D $3x + 2y - 7 = 0$

Classified Practice

Triangles and Lines

Q161 Written

Find the line through $(-2, 5)$ perpendicular to $3x + 5y + 1 = 0$.

Q162 Written

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$.

Classified Practice

Triangles and Lines

Q163 **Written**

Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$.

Q164 Written

Find the equation of the line through $(-1,3)$ perpendicular to $5x - 2y + 3 = 0$.

Classified Practice

Triangles and Lines

Q165 Written

Find the equation of the line through $(-2, -5)$ perpendicular to $3x - 2y - 7 = 0$.

Q166 MCQ**Find the line through $(-2,5)$ perpendicular to $3x + 5y + 1 = 0$:****Choose the correct option.**

A $5x - 3y + 25 + 1 = 0$

B $5x - 3y - 25 = 0$

C $2(5x - 3y + 25) + 1 = 0$

D $5x - 3y + 25 = 0$

Classified Practice

Triangles and Lines

Q167 MCQ

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$:**Choose the correct option.**

A $3x - 2y - 4 + 1 = 0$

B $3x + 2y - 4 = 0$

C $2(3x - 2y - 4) + 1 = 0$

D $3x - 2y - 4 = 0$

Q168 MCQ**Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$:****Choose the correct option.**

A $2(x - 2y - 12) + 1 = 0$

B $x - 2y - 12 = 0$

C $x - 2y - 12 + 1 = 0$

D $x + 2y - 12 = 0$

Classified Practice

Triangles and Lines

Q169 MCQ

Find the equation of the line through $(-1,3)$ perpendicular to $5x - 2y + 3 = 0$:

Choose the correct option.

A $2x + 5y - 13 = 0$

B $2x + 5y - 13 + 1 = 0$

C $2x - 5y - 13 = 0$

D $2(2x + 5y - 13) + 1 = 0$

Q170 MCQ

Find the equation of the line through $(-2, -5)$ perpendicular to $3x - 2y - 7 = 0$:

Choose the correct option.

A $2x + 3y + 19 + 1 = 0$

B $2x - 3y + 19 = 0$

C $2(2x + 3y + 19) + 1 = 0$

D $2x + 3y + 19 = 0$

Classified Practice

Triangles and Lines

Q171 Challenge

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k .

Classified Practice

Triangles and Lines

Q172 Challenge

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k :

Choose the correct option.

- A $m/k = 1$ ($k = 2, m = 2$)
 - B $m/k = 3$ ($k = 2, m = 6$)
 - C $m/k = -2$ ($k = 1, m = -2$)
 - D $m/k = 2$ ($k = 3, m = 6$)
-
-
-

Classified Practice

Triangles and Lines

Q173 **Written**

For A (5, 11), B (1, -1), find the equation of the perpendicular bisector of AB.

Q174 Written

For A $(-1, 2)$, B $(3, -4)$, find the equation of the perpendicular bisector of AB.

Classified Practice

Triangles and Lines

Q175 **Written****Find the perpendicular bisector of segment A $(-2, 3)$, B $(2, 1)$.**

Q176 Written**Find the perpendicular bisector of segment A $(-1, 2)$, B $(5, -2)$.**

Classified Practice

Triangles and Lines

Q177 **Written****Find the perpendicular bisector of segment A (0, 6), B (4, 4).**

Q178 MCQ**For A (5, 11), B (1, -1), find the equation of the perpendicular bisector of AB:****Choose the correct option.**

A $x + 3y - 18 = 0$

B $x + 3y - 18 + 1 = 0$

C $x - 3y - 18 = 0$

D $2(x + 3y - 18) + 1 = 0$

Classified Practice

Triangles and Lines

Q179 MCQ

For A $(-1, 2)$, B $(3, -4)$, find the equation of the perpendicular bisector of AB:

Choose the correct option.

A $2x + 3y - 5 = 0$

B $2(2x - 3y - 5) + 1 = 0$

C $2x - 3y - 5 = 0$

D $2x - 3y - 5 + 1 = 0$

Q180 MCQ**Find the perpendicular bisector of segment A $(-2, 3)$, B $(2, 1)$:****Choose the correct option.**

A $2x - y - 2 = 0$

B $2(2x - y + 2) + 1 = 0$

C $2x - y + 2 = 0$

D $2x - y + 2 + 1 = 0$

Classified Practice

Triangles and Lines

Q181 MCQ

Find the perpendicular bisector of segment A (−1, 2), B (5, −2):**Choose the correct option.**

A $3x - 2y - 6 + 1 = 0$

B $3x + 2y - 6 = 0$

C $2(3x - 2y - 6) + 1 = 0$

D $3x - 2y - 6 = 0$

Q182 MCQ**Find the perpendicular bisector of segment A (0, 6), B (4, 4):****Choose the correct option.**

A $2x - y + 1 = 0$

B $2x - y + 1 + 1 = 0$

C $2x - y - 1 = 0$

D $2(2x - y + 1) + 1 = 0$

Classified Practice

Triangles and Lines

Q183 **Written**

Find the point B symmetric to A $(-4, 3)$ with respect to $3x - y + 5 = 0$.

Q184 Written**Find the point B symmetric to A (0, 4) with respect to $2x - y - 1 = 0$.**

Classified Practice

Triangles and Lines

Q185 **Written****Find the point B symmetric to A (1, 2) with respect to $x + 2y - 10 = 0$.**

Q186 Written**Find the point B symmetric to A (4, -1) with respect to $4x - 2y - 3 = 0$.**

Classified Practice

Triangles and Lines

Q187 MCQ

Find the point B symmetric to A $(-4, 3)$ with respect to $3x - y + 5 = 0$:

Choose the correct option.

A $B = (2, 1)$

B $B = (1, 2)$

C $B = (-2, 1)$

D $B = (2, -1)$

Q188 MCQ**Find the point B symmetric to A (0, 4) with respect to $2x - y - 1 = 0$:****Choose the correct option.****A** $B = (4, 2)$ **B** $B = (2, 4)$ **C** $B = (-4, 2)$ **D** $B = (4, -2)$

Classified Practice

Triangles and Lines

Q189 MCQ

Find the point B symmetric to A (1, 2) with respect to $x + 2y - 10 = 0$:

Choose the correct option.

A $B = (-3, 6)$

B $B = (3, -6)$

C $B = (3, 6)$

D $B = (6, 3)$

Q190 MCQ**Find the point B symmetric to A (4, -1) with respect to $4x - 2y - 3 = 0$:****Choose the correct option.**

A $B = (2, 2)$

B $B = (-2, -2)$

C $B = (-2, 2)$

D $B = (2, -2)$

Classified Practice

Triangles and Lines

Q191 **Challenge****Reflect A (5, 0) across the targeting line $y = 2x$. Find the receiver point B.**

Classified Practice

Triangles and Lines

Q192 Challenge

Reflect A (5, 0) across the targeting line $y = 2x$. Find the receiver point B:

Choose the correct option.

A $B = (3, 4)$

B $B = (-3, -4)$

C $B = (-3, 4)$

D $B = (4, -3)$

Classified Practice

Triangles and Lines

Q193 Written

Find the distance from $(-2,1)$ to $3x - 2y + 5 = 0$.

Q194 Written**Find the distance from (3,-2) to $x + 2y - 4 = 0$.**

Classified Practice

Triangles and Lines

Q195 **Written****Find the perpendicular distance from $(-3,2)$ to $2x - 3y + 6 = 0$.**

Q196 Written**Find the perpendicular distance from $(-3,7)$ to $12x - 5y = 7$.**

Classified Practice

Triangles and Lines

Q197 **Written****Find the perpendicular distance from $(-8,1)$ to $y = 2x - 3$.**

Q198 Written**Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$.**

Classified Practice

Triangles and Lines

Q199 Written

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$.

Q200 Written**Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$.**

Classified Practice

Triangles and Lines

Q201 **Written****Find the perpendicular distance from (1,2) to $y = 3x + 1$.**

Q202 MCQ

Find the distance from $(-2,1)$ to $3x - 2y + 5 = 0$:**Choose the correct option.**

A $3/\sqrt{13} = 3\sqrt{13}/13$

B $3 \cdot \sqrt{13}/13 + 1$

C $-1 + 3 \cdot \sqrt{13}/13$

D $-3 \cdot \sqrt{13}/13$

Classified Practice

Triangles and Lines

Q203 MCQ

Find the distance from (3,-2) to $x + 2y - 4 = 0$:**Choose the correct option.**

A $1 + \sqrt{5}$

B $-1 + \sqrt{5}$

C $-\sqrt{5}$

D $\sqrt{5}$

Q204 MCQ

Find the perpendicular distance from $(-3,2)$ to $2x - 3y + 6 = 0$:**Choose the correct option.**

A $1 + 6 \cdot \sqrt{13}/13$

B $-1 + 6 \cdot \sqrt{13}/13$

C $-6 \cdot \sqrt{13}/13$

D $6/\sqrt{13} = 6\sqrt{13}/13$

Classified Practice

Triangles and Lines

Q205 MCQ

Find the perpendicular distance from $(-3,7)$ to $12x - 5y = 7$:**Choose the correct option.**A -6 B 6 C 7 D 5

Q206 MCQ**Find the perpendicular distance from $(-8,1)$ to $y = 2x - 3$:****Choose the correct option.**

A $4\sqrt{5} + 1$

B $4\sqrt{5} - 1$

C 0

D $4\sqrt{5}$

Classified Practice

Triangles and Lines

Q207 MCQ

Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$:**Choose the correct option.**

A 2

B 0

C -1

D 1

Q208 MCQ

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$:**Choose the correct option.**

- A $\sqrt{13}$
 - B $1 + \sqrt{13}$
 - C $-1 + \sqrt{13}$
 - D $-\sqrt{13}$
-
-
-

Classified Practice

Triangles and Lines

Q209 MCQ

Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$:**Choose the correct option.**

A -10

B 10

C 11

D 9

Q210 MCQ**Find the perpendicular distance from (1,2) to $y = 3x + 1$:****Choose the correct option.**

- A $\sqrt{10}/5 + 1$
 - B $-1 + \sqrt{10}/5$
 - C $-\sqrt{10}/5$
 - D $\sqrt{10}/5$
-
-
-

Classified Practice

Triangles and Lines

Q211 Written

Find the perpendicular length from the origin to $3x - y = 5$.

Q212 Written**Find the perpendicular length from $(-1,5)$ to $y = 3x - 2$.**

Classified Practice

Triangles and Lines

Q213 MCQ

Find the perpendicular length from the origin to $3x - y = 5$:**Choose the correct option.**

- A $1 + \sqrt{10}/2$
- B $-1 + \sqrt{10}/2$
- C $-\sqrt{10}/2$
- D $\sqrt{10}/2$

Q214 MCQ

Find the perpendicular length from $(-1,5)$ to $y = 3x - 2$:**Choose the correct option.**

A $-\sqrt{10}$

B $\sqrt{10}$

C $1 + \sqrt{10}$

D $-1 + \sqrt{10}$

Classified Practice

Triangles and Lines

Q215 **Challenge****Find the shortest distance from P (−1, 5) to the radiation line $y = x/3 + 2$.**

Classified Practice

Triangles and Lines

Q216 Challenge

Find the shortest distance from P $(-1, 5)$ to the radiation line $y = x/3 + 2$:

Choose the correct option.

- A $\sqrt{10}$
- B $1 + \sqrt{10}$
- C $-1 + \sqrt{10}$
- D $-\sqrt{10}$

Classified Practice

Triangles and Lines

Q217 **Written****Find the area of triangle ABC with A (1, 1), B (3, 7), C (−3, −1).**

Q218 Written

Find the area for A (1, 5), B (−3, −3), C (3, −6).

Classified Practice

Triangles and Lines

Q219 **Written****Find the area for A $(-2, 1)$, B $(2, -1)$, C $(0, 4)$.**

Q220 Written**Find the area for A (0, -6), B (2, -2), C (1, 5).**

Classified Practice

Triangles and Lines

Q221 **Written****Find the area for A $(-2, 4)$, B $(-3, -5)$, C $(5, -1)$.**

Q222 Written**Find the area for A (2, 5), B (0, 1), C (4, 2).**

Classified Practice

Triangles and Lines

Q223 MCQ

Find the area of triangle ABC with A (1, 1), B (3, 7), C (−3, −1):**Choose the correct option.**

A 11

B 9

C −10

D 10

Q224 MCQ

Find the area for A (1, 5), B (−3, −3), C (3, −6):**Choose the correct option.**

A 29

B −30

C 30

D 31

Classified Practice

Triangles and Lines

Q225 MCQ

Find the area for A $(-2, 1)$, B $(2, -1)$, C $(0, 4)$:**Choose the correct option.**

A 8

B 9

C 7

D -8

Q226 MCQ**Find the area for A (0, -6), B (2, -2), C (1, 5):****Choose the correct option.****A** -9**B** 9**C** 10**D** 8

Classified Practice

Triangles and Lines

Q227 MCQ

Find the area for A $(-2, 4)$, B $(-3, -5)$, C $(5, -1)$:**Choose the correct option.**A -34 B 34 C 35 D 33

Q228 MCQ

Find the area for A (2, 5), B (0, 1), C (4, 2):**Choose the correct option.**

A 8

B 6

C -7

D 7

Classified Practice

Triangles and Lines

Q229 **Challenge****Find the area of the wound triangle A $(-2, 1)$, B $(1, 5)$, C $(-21, -21)$.**

Classified Practice

Triangles and Lines

Q230 Challenge

Find the area of the wound triangle A $(-2, 1)$, B $(1, 5)$, C $(-21, -21)$:

Choose the correct option.

A 6

B 4

C -5

D 5

Classified Practice

Triangles and Lines

Q231 Written

The line $(k - 4)x + (k + 2)y + k + 5 = 0$ passes through a fixed point for every k . Find it.

Q232 Written

The line $(2k + 1)x - (k + 2)y + 7k + 8 = 0$ passes through a fixed point for every k . Find it.

Classified Practice

Triangles and Lines

Q233 Written

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point for every k . Find it.

Q234 Written

The line $(k + 3)x - (2k + 1)y + k - 2 = 0$ passes through a fixed point for every k . Find it.

Classified Practice

Triangles and Lines

Q235 Written

The line $(k + 3)x + (3k + 2)y + 7 = 0$ passes through a fixed point for every k . Find it.

Q236 MCQ

The line $(k - 4)x + (k + 2)y + k + 5 = 0$ passes through a fixed point for every k . Find it:

Choose the correct option.

A $(-3/2, 1/2)$

B $(-1/2, -3/2)$

C $(1/2, 3/2)$

D $(1/2, -3/2)$

Classified Practice

Triangles and Lines

Q237 MCQ

The line $(2k + 1)x - (k + 2)y + 7k + 8 = 0$ passes through a fixed point for every k . Find it:

Choose the correct option.

A $(-2, -3)$

B $(-2, 3)$

C $(3, -2)$

D $(2, 3)$

Q238 MCQ

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point for every k . Find it:

Choose the correct option.

A $(-3, 3)$

B $(-3, -3)$

C $(3, 3)$

D $(3, -3)$

Classified Practice

Triangles and Lines

Q239 MCQ

The line $(k + 3)x - (2k + 1)y + k - 2 = 0$ passes through a fixed point for every k . Find it:

Choose the correct option.

A $(-1, 1)$

B $(1, -1)$

C 0

D $(1, 1)$

Q240 MCQ

The line $(k + 3)x + (3k + 2)y + 7 = 0$ passes through a fixed point for every k . Find it:

Choose the correct option.

A $(-3, 1)$

B $(1, -3)$

C $(3, 1)$

D $(-3, -1)$

Classified Practice

Triangles and Lines

Q241 **Challenge**

The line $-(k - 2)x + (k - 3)y = -3k + 5$ passes through a fixed point for every k . Find it.

Classified Practice

Triangles and Lines

Q242 Challenge

The line $-(k - 2)x + (k - 3)y = -3k + 5$ passes through a fixed point for every k . Find it:

Choose the correct option.

A (4, 1)

B (1, 4)

C (-4, 1)

D (4, -1)

Classified Practice

Triangles and Lines

Q243 Written

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a .

Q244 Written

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a .

Classified Practice

Triangles and Lines

Q245 **Written**

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a .

Q246 Written

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a .

Classified Practice

Triangles and Lines

Q247 MCQ

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a:

Choose the correct option.

A $a = -4, -3, \text{ or } 6$

B $a = -3, -3, \text{ or } 6$

C $a = -4, -2, \text{ or } 6$

D $a = -4, -3, \text{ or } 7$

Q248 MCQ

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a:

Choose the correct option.

A $a = -1, -1/2, \text{ or } 1$

B $a = 0, -1/2, \text{ or } 1$

C $a = -1, 1/2, \text{ or } 1$

D $a = -1, -1/2, \text{ or } 2$

Classified Practice

Triangles and Lines

Q249 MCQ

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a :

Choose the correct option.

A $a = 11/10$, $5/2$, or $-9/2$

B $a = 11/10$, $3/2$, or $-7/2$

C $a = 11/10$, $3/2$, or $-9/2$

D $a = 21/10$, $3/2$, or $-9/2$

Q250 MCQ

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a:

Choose the correct option.

A $a = 2, -3/2, \text{ or } 2$

B $a = 1, -1/2, \text{ or } 2$

C $a = 1, -3/2, \text{ or } 3$

D $a = 1, -3/2, \text{ or } 2$

Classified Practice

Triangles and Lines

Q251 Challenge

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m .

Classified Practice

Triangles and Lines

Q252 Challenge

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m :

Choose the correct option.

A 15

B 16

C 14

D -15

Classified Practice

Triangles and Lines

Quiz MCQ 1 **Concept Quiz · MCQ**

Determine the shape of triangle A (0, 0), B (3, 0), C (0, 4):

Choose the correct option.

A *isosceles (not right – angled)*

B *scalene (not right – angled)*

C *scalene right – angled*

D *equilateral*

Quiz MCQ 2 Concept Quiz · MCQ

Find the equation of the line through (2,-1) with gradient 3:

Choose the correct option.

A 0

B $y = 3x - 7$

C $y = -3x - 7$

D $y = 3x - 7 + 1$

Classified Practice

Triangles and Lines

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)**Find the perpendicular distance from (1,2) to $3x + 4y - 10 = 0$:****Choose the correct option.**A $6/5$ B $-4/5$ C $-1/5$ D $1/5$

Quiz MCQ 4 Concept Quiz · MCQ

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point. Find it:

Choose the correct option.

A $(-3, -3)$

B $(3, 3)$

C $(3, -3)$

D $(-3, 3)$

Classified Practice

Triangles and Lines

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the area of triangle A (0, 0), B (4, 0), C (0, 3).

Quiz WRITTEN 6 Concept Quiz · Written

Find the perpendicular bisector of the segment joining A (1, 2) and B (5, 4).

Classified Practice

Triangles and Lines

Quiz WRITTEN 7 Concept Quiz · Written

Find the equation of the line through (1,2) perpendicular to $y = 3x - 1$.

Quiz WRITTEN 8 Concept Quiz · Written

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a .

Answer key

Use the question number shown on each page.

Q001 <i>equilateral</i>	Q027 $x = -3$	Q053 $x = -5$	Q079 C
Q002 <i>equilateral</i>	Q028 $x = 3$	Q054 $y = 3$	Q080 D
Q003 D	Q029 $x = 2$	Q055 $y = -2$	Q081 B
Q004 C	Q030 $x = 4$	Q056 $y = 6$	Q082 D
Q005 <i>isosceles right – angled; right angle at A</i>	Q031 $x = -6$	Q057 $y = 4x - 2$	Q083 $y = 2$
Q006 <i>isosceles right – angled; right angle at A</i>	Q032 B	Q058 $y = 2x - 6$	Q084 D
Q007 B	Q033 C	Q059 B	Q085 $a = 7$
Q008 D	Q034 D	Q060 C	Q086 $a = -1$
Q009 <i>scalene right – angled; right angle at A</i>	Q035 D	Q061 D	Q087 $a = 1$
Q010 D	Q036 B	Q062 A	Q088 $a = 7$
Q011 <i>isosceles right – angled; right angle at B</i>	Q037 D	Q063 B	Q089 $a = 5$
Q012 <i>isosceles right – angled; right angle at B</i>	Q038 B	Q064 D	Q090 D
Q013 <i>isosceles right – angled; right angle at C</i>	Q039 C	Q065 C	Q091 D
Q014 B	Q040 D	Q066 A	Q092 C
Q015 C	Q041 D	Q067 C	Q093 D
Q016 B	Q042 B	Q068 D	Q094 D
Q017 $k = 2$	Q043 D	Q069 D	Q095 $x + y - 2 = 0$
Q018 C	Q044 A	Q070 C	Q096 $x + 3y - 23 = 0$
Q019 $y = -4$	Q045 $x = -3$	Q071 D	Q097 $x + y - 1 = 0$
Q020 $x = -3$	Q046 $y = 3$	Q072 D	Q098 $2x - 5y + 1 = 0$
Q021 $y = 6$	Q047 $y = x/3 - 4/3$ (equivalently $x - 3y - 4 = 0$)	Q073 $y = 3x - 7$	Q099 B
Q022 $x = 3$	Q048 $y = -2$	Q074 $y = 2x + 10$	Q100 A
Q023 $y = 5$	Q049 $x = -4$	Q075 $y = -x + 4$	Q101 B
Q024 $y = 1$	Q050 $7x + 2y + 26 = 0$	Q076 $y = x - 5$	Q102 D
Q025 $y = -7$	Q051 $x = 4$	Q077 $y = -2x + 3$	Q103 $3x + 13y = 0$
Q026 $x = 1$	Q052 $x = -1$	Q078 D	Q104 C

Q105 $m = 2$	Q137 C	Q169 A	Q201 $\sqrt{10}/5$
Q106 $m = -3$	Q138 C	Q170 D	Q202 A
Q107 $m = 1/3$	Q139 D	Q171 $m/k = 2 (k = 3, m = 6)$	Q203 D
Q108 $m = 2$	Q140 A	Q172 D	Q204 D
Q109 $m = -1/3$	Q141 D	Q173 $x + 3y - 18 = 0$	Q205 B
Q110 $m = 3$	Q142 C	Q174 $2x - 3y - 5 = 0$	Q206 D
Q111 $m = 4$	Q143 C	Q175 $2x - y + 2 = 0$	Q207 D
Q112 $m = -4$	Q144 A	Q176 $3x - 2y - 6 = 0$	Q208 A
Q113 $m = 3$	Q145 D	Q177 $2x - y + 1 = 0$	Q209 B
Q114 $m = -4$	Q146 A	Q178 A	Q210 D
Q115 $m = -1/4$	Q147 B	Q179 C	Q211 $\sqrt{10}/2$
Q116 $m = 3$	Q148 D	Q180 C	Q212 $\sqrt{10}$
Q117 $m = -3$	Q149 B	Q181 D	Q213 D
Q118 $m = 1$	Q150 B	Q182 A	Q214 B
Q119 $m = -3$	Q151 $3x - 2y - 14 = 0$	Q183 $B = (2, 1)$	Q215 $\sqrt{10}$
Q120 $m = 1$	Q152 $2x + y + 1 = 0$	Q184 $B = (4, 2)$	Q216 A
Q121 $m = -1/3$	Q153 $2x - y - 4 = 0$	Q185 $B = (3, 6)$	Q217 10
Q122 $m = 2$	Q154 $2x + 3y - 4 = 0$	Q186 $B = (-2, 2)$	Q218 30
Q123 $m = -1$	Q155 $3x + 2y - 7 = 0$	Q187 A	Q219 8
Q124 $m = 3/2$	Q156 C	Q188 A	Q220 9
Q125 $m = -1$	Q157 D	Q189 C	Q221 34
Q126 $m = -2/3$	Q158 C	Q190 C	Q222 7
Q127 $m = -1/2$	Q159 D	Q191 $B = (-3, 4)$	Q223 D
Q128 D	Q160 D	Q192 C	Q224 C
Q129 C	Q161 $5x - 3y + 25 = 0$	Q193 $3/\sqrt{13} = 3\sqrt{13}/13$	Q225 A
Q130 A	Q162 $3x - 2y - 4 = 0$	Q194 $\sqrt{5}$	Q226 B
Q131 B	Q163 $x - 2y - 12 = 0$	Q195 $6/\sqrt{13} = 6\sqrt{13}/13$	Q227 B
Q132 A	Q164 $2x + 5y - 13 = 0$	Q196 6	Q228 D
Q133 A	Q165 $2x + 3y + 19 = 0$	Q197 $4\sqrt{5}$	Q229 5
Q134 B	Q166 D	Q198 1	Q230 D
Q135 D	Q167 D	Q199 $\sqrt{13}$	Q231 $(1/2, -3/2)$
Q136 A	Q168 B	Q200 10	Q232 $(-2, 3)$

Q233 (3, -3)

Q234 (1, 1)

Q235 (-3, 1)

Q236 D

Q237 B

Q238 D

Q239 D

Q240 A

Q241 (4, 1)

Q242 A

Q243 $a = -4, -3, \text{ or } 6$ Q244 $a = -1, -1/2, \text{ or } 1$ Q245 $a = 11/10, 3/2, \text{ or } -9/2$ Q246 $a = 1, -3/2, \text{ or } 2$

Q247 A

Q248 A

Q249 C

Q250 D

Q251 15

Q252 A

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 D

Quiz MCQ 4 C

Quiz WRITTEN 5 6

Quiz WRITTEN 6 $y = -2x + 9$ Quiz WRITTEN 7 $y = -x/3 + 7/3$ Quiz WRITTEN 8 $a = -4, -3, \text{ or } 6$